POPULATION, URBANIZATION, INDUSTRIALIZATION AND RESOURCE USE

From human development to resource pressure

More people demand more resources, infrastructure, mobility and energy.

From Earth Systems to Human Pressure

EARTH SYSTEM → **Atmosphere ↔ Hydrosphere ↔ Geosphere ↔ Biosphere**

Human dependence → Water • Food • Energy • Land • Materials

Human activity → Population • Urbanization • Industry • Infrastructure

ENVIRONMENTAL PRESSURE

How large has this pressure become — and how quickly is it increasing?

The Human Population: A Century of Change

POPULATION

Global population has increased more than fourfold since the mid-1920s.

1.91 B	2.49 B	6.14 B	~8.2 B
1925	1950	2000	2024/25

≈ 4.3× increase in about a century

More people → more food, water, housing, mobility, energy and infrastructure demand.

Source: UN World Population Prospects / historical UN estimates

India's Population: From ~250 Million to ~1.46 Billion

1921	1951	1971	1991	2011	2025*
251 M	361 M	548 M	846 M	1.21 B	~1.46 B

≈ 5.8× increase since 1921

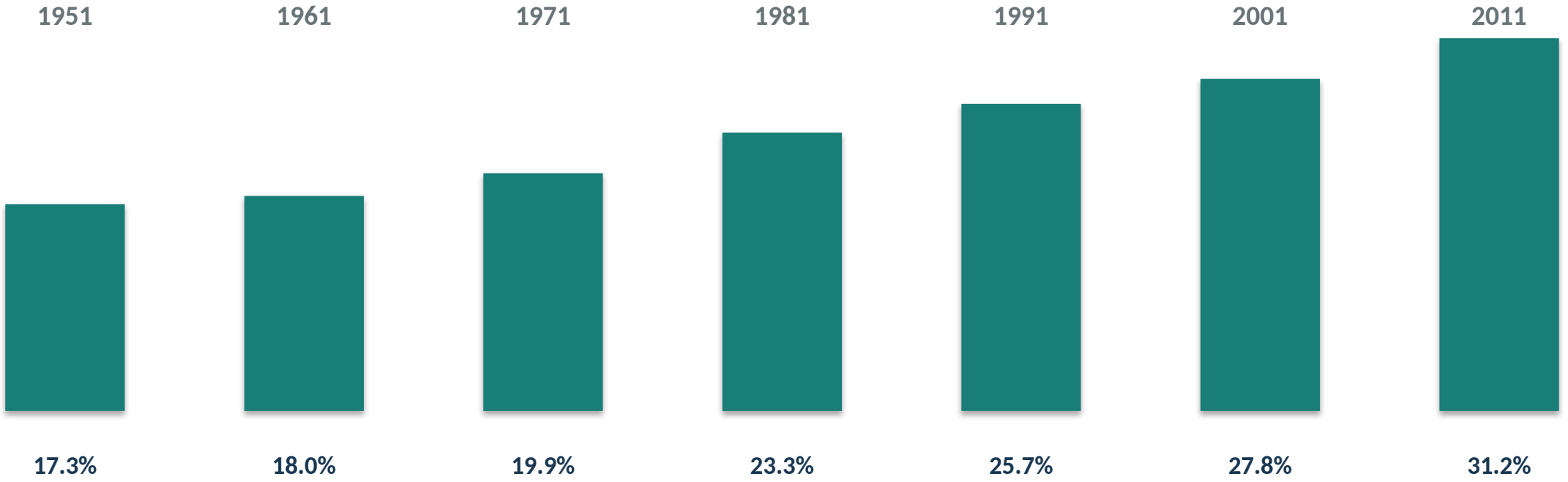
Population growth creates infrastructure demand — but environmental pressure also depends on consumption per person and technology.

*2025 is an estimate; Census 2011 remains the latest completed population census.

Sources: Census of India; UN World Population Prospects 2024

The scale of change is even more striking in India.

India: From a Predominantly Rural Society to an Urbanizing Nation



~17% → ~31% of Indians lived in urban areas

2011 is the latest completed Census benchmark; later values should be labelled estimates.

Urbanization: More Than a Percentage

The absolute number of urban residents has grown enormously.

~62 M

Urban population, 1951

~286 M

Urban population, 2001

~377 M

Urban population, 2011

What does an additional ~300+ million urban residents mean?

Housing . Water . Sewerage . Roads . Electricity . Public transport
Waste management . Drainage

Civil Engineering translates demographic growth into physical infrastructure.

Source: Census of India / World Bank historical urbanization series

Urban Concentration: Density Changes the Engineering Problem

9,340

Delhi density, 2001 persons/km²

11,297

Delhi density, 2011 persons/km²

~21%

increase in one decade

Higher density can mean:

Greater demand per unit area

- Water supply
- Sewerage
- Roads & transit
- Drainage
- Solid waste systems
- Open space

Source: Census of India 2011

From Population Growth to the Built Environment

Demographic growth becomes physical infrastructure.



The built environment is the physical expression of population and economic demand.

Civil Engineering sits at the interface between demographic growth and natural resource consumption.

Mobility: The Transportation Explosion

India's transport infrastructure and vehicle fleet have expanded dramatically.

~4.0 lakh km

Road network, 1951

~63.6 lakh km

Road network, 2020

~16×

road-network expansion

Registered motor vehicles

~55 M (2001) → 326+ M (2020) ≈ 6× increase

More mobility → more roads, vehicles, fuel/electricity, parking, materials and urban space.

Source: Ministry of Road Transport & Highways, Basic Road Statistics / Road Transport Year Book

Industrialization: From Development to Resource Demand

Industry converts large quantities of natural resources into economic value.

RAW MATERIALS **ENERGY** **WATER** **LAND** **TRANSPORT**

↓

INDUSTRIAL ACTIVITY

↓

PRODUCTS + EMPLOYMENT + ECONOMIC VALUE

Alongside

WASTE + EMISSIONS + RESOURCE DEPLETION

Industrialization increases the scale and speed at which resources are transformed.

India's Material Scale: Steel as an Indicator

Modern development requires enormous quantities of industrial materials.

165.6 Mt

India crude steel production, 2025

#2

India's global rank, 2025

Steel supports:

Buildings • bridges • reinforcement
railways • vehicles • industrial equipment

This is only ONE material stream.

Add cement + aggregates + sand + bricks + glass +
Aluminium + timber + polymers + energy …

Source: World Steel Association, 2025 crude steel production

Civil Engineering: The Physical Engine of Development

Population and economic growth become infrastructure through engineering.

Housing

Buildings + land + cement + steel + water

Mobility

Roads + rail + bridges + transport systems

Water

Sources + treatment + distribution

Sanitation

Sewerage + treatment + reuse

Stormwater

Drainage + flood management

Waste

Collection + processing + disposal

Civil Engineering converts societal demand into physical infrastructure — and therefore strongly influences resource consumption.

What Resources Does Development Consume?

Four resource groups are especially important for Civil Engineering.

WATER

Domestic • agriculture • industry • construction

LAND

Housing • infrastructure • industry • agriculture

MINERALS

Iron • limestone • sand • aggregates • aluminium

ENERGY

Coal • oil • gas • electricity • renewables

Civil Engineering directly depends on land + water + minerals + energy.

Water: A Resource Under Pressure

India has a large population share but a much smaller share of global water resources.

~18%

of world's population

~4%

of global water resources

~50%

fall in per-capita water availability since 1970

Current pressure is intensified by:

Seasonal rainfall
Groundwater dependence
Urban growth
Industrial demand
Agricultural demand
Climate variability

Civil Engineering challenge: provide reliable services while reducing losses, over-extraction and inefficient use.

Source: World Bank, India water brief

Land and Minerals: The Hidden Material Footprint

Infrastructure growth is physically built from extracted resources.

LAND TRANSFORMATION

Natural / agricultural / open land

↓

Buildings + roads + utilities

MINERAL EXTRACTION

Iron ore • limestone • sand • aggregates • clay

↓

Steel + cement + concrete + masonry

Construction is therefore a major material transformation process.

Quarry → aggregate → concrete → building → demolition → C&D waste

Key question: Can infrastructure growth continue with a purely linear extract–use–discard model?

Energy Demand: The Hidden Driver

Every modern service depends on energy — including water and waste infrastructure.

BUILDINGS	TRANSPORT	INDUSTRY	WATER & WASTEWATER
construction + operation	people + goods movement	manufacturing + processing	pumping + treatment

INDIA'S ENERGY DEMAND

↑ Urbanization + ↑ Manufacturing + ↑ Built environment + ↑ Mobility

Engineering opportunity: efficiency in design, materials, operation and maintenance can reduce resource intensity.

Source: International Energy Agency — India energy analyses

The Conventional Linear Resource Model

Business-as-usual development follows a largely one-way material flow.

EXTRACT → PROCESS → MANUFACTURE → BUILD → USE → DEMOLISH / DISCARD

At every stage → Materials • Energy • Water • Land

Outputs → Products + Waste + Emissions + Environmental pressure

Civil Engineering example: quarry → aggregate → concrete → building → demolition → C&D waste

The Population–Urbanization–Industrialization–Resource Nexus

The drivers are connected — and so are their consequences.



RESOURCE EXTRACTION

↓

ENVIRONMENTAL PRESSURE

Modified by: consumption pattern • technology • efficiency • planning • governance

Can We Continue With the Same Resource-Use Pattern?

The challenge is not development itself — it is the resource intensity of development.

BUSINESS AS USUAL

More people + more consumption + resource-intensive technology + linear material use

↓

RAPIDLY INCREASING RESOURCE PRESSURE

ENGINEERING RESPONSE

Resource efficiency
Material substitution
Water efficiency
Energy efficiency
Public transport
Compact planning
Reuse & recycling
Circular material flows

The engineer influences not only how much infrastructure is built, but how much resource is required across its life cycle.

What Have the Numbers Shown Us?

A quantitative summary before we examine resource limits.

WORLD POPULATION

~1.9 B → ~8.2 B

~4.3×

INDIA

251 M → ~1.46 B

~5.8×

INDIA URBAN SHARE

~17% → 31%

1951 → 2011

ROAD NETWORK

~4.0 → 63.6 lakh km

1951 → 2020

VEHICLES

~55 M → 326+ M

2001 → 2020

WATER

~18% pop. / ~4% water

Strong pressure

STEEL

165.6 Mt

India, 2025

NEXT QUESTION

What happens when human demand begins to exceed the regenerative capacity and availability of natural resources?

→ Presentation 3: Resource Depletion, Carrying Capacity and Sustainability Challenges